Dreamer from Pixels: Midterm Baseline and Reproduction Plan

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Abstract

We study visual control with Dreamer V1 in the DeepMind Control Suite. We use Julius Frost's PyTorch implementation, fix the source commit, and test the pipeline on `cartpole_swingup` before extending to other tasks. Our completed 100k baseline uses 64 x 64 RGB pixels, action repeat 2, one training seed, and 100,000 environment steps; its final policy reached 181.701 ± 27.571 on ten evaluation episodes. We then ran an independent seed-0 extension that sampled 330,732 environment steps and 16,500 updates before the wall-clock cutoff. The reported 322.084 ± 9.687 evaluation uses the saved `params.pkl` snapshot at `itr=164999`, representing 165,000 decisions or 330,000 environment steps in the checkpoint payload. These are preliminary single-seed results, not a full paper reproduction. We still need matched training seeds and the two proposal experiments.

1 What We Want to Study

Visual control requires an agent to learn both a useful representation and a policy from raw images. Dreamer addresses this problem with a learned latent dynamics model and policy improvement inside imagined trajectories. Our main goal is to reproduce and understand this Dreamer V1 pipeline from pixels. We start with `cartpole_swingup`, then plan to evaluate `walker_walk` and `ball_in_cup_catch` after the baseline is stable.

We evaluate control with episodic return and learning curves. We do not use classification accuracy because the task is continuous control. We use a single training seed for the current pilot and keep the evaluation protocol separate from the training interaction. The current report is an initial baseline report, not a claim that we have matched the paper's full experimental study.

2 Background and Related Work

PlaNet introduced recurrent latent dynamics for planning from pixels [4]. Dreamer follows the same general direction but replaces repeated online planning with value learning through latent imagination: it rolls the actor forward in the learned model and sends return gradients back through the imagined dynamics [3]. We use Dreamer V1 as the baseline and cite PlaNet to explain the predecessor relationship; we do not combine their results or implementations.

We use Julius Frost's PyTorch implementation of Dreamer V1 at commit `ccea6ae4a397a94c328891bd78e81d52dd156cb6`, with the pinned `rlpyt` submodule at `f04f23db1eb7b5915d88401fca67869968a07a37` [1]. The DeepMind Control Suite provides the simulated continuous-control task and reward contract [5]. We preserve the upstream MIT license and record our wrapper and measurement changes separately. The original Google Research

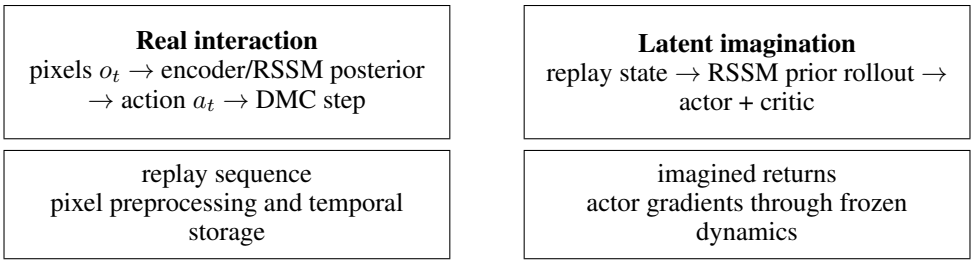


Figure 1: Our simplified Dreamer data flow. Real interaction supplies image sequences to replay. Training starts from replay states and uses the learned RSSM for latent imagination; the critic supplies return targets and actor gradients pass through the imagined dynamics.

Table 1: Effective configuration used by the formal runs. The overnight run used the same settings and stopped at its wall-clock limit.

Task / observation	cartpole_swingup, camera 0, 64 x 64 RGB
Training seed	0
Decision / environment budget	50,000 / 100,000 baseline; 250,000 / 500,000 target for overnight run
Action repeat / episode limit	2 / 500 decisions (1,000 env steps)
Prefill	1,000 decisions (2,000 env steps)
Training schedule	every 1,000 decisions, 100 updates
Batch size / length	16 / 50
Replay capacity	60,000 decision transitions
Precision / `use_pcont`	32-bit / False
Evaluation environment seeds	1000–1009

repository is retained as a historical Dreamer V1 software reference, not as a second implementation used in this run [2].

3 Method

3.1 Dreamer path

The agent stores image sequences generated by real environment interaction. The world model encodes observations and infers a recurrent stochastic state. The actor and critic then operate on latent states imagined from replay states. Figure 1 summarizes the two paths.

The environment wrapper renders camera 0 as a 64 x 64 RGB image and stores it as a uint8 tensor in channel-first format. Before the encoder, the implementation converts pixels from $[0, 255]$ to $[-0.5, 0.5]$. One action is repeated for two underlying environment steps. Thus, 50,000 policy decisions correspond to 100,000 environment steps. Episodes contain 500 decisions, or 1,000 environment steps. The replay buffer stores temporal sequences, and the training batch has 16 sequences of length 50.

3.2 Objectives used in this run

The world model learns an observation reconstruction distribution, a reward distribution, and an RSSM prior/posterior agreement term. The implementation also supports a continuation-probability head, but our formal run uses `use_pcont=False`. Therefore, the continuation head is not constructed, its loss is not included in the world-model objective, and the imagined-return calculation uses the fixed discount factor instead. The logged `pcont_loss` field is a zero placeholder when this option is disabled; it is not evidence that a continuation loss was trained.

The actor is trained on returns imagined from posterior states, while the critic predicts those imagined returns. The main checks for the debug gate were finite losses and gradients, nonzero changes to all three parameter groups, and loading the saved model in a new process. We do not claim full optimizer/replay/environment-state continuation from the exported checkpoint.

Table 2: Independent iterations. Budget, replay, and initial-checkpoint handling changed together; this is not a one-factor ablation. The overnight run stopped at 330,732 environment steps.

	6k pilot	100k baseline	overnight run
Decision / environment steps	3,000 / 6,000	50,000 / 100,000	165,000 / 330,000 snapshot; 165,366 / 330,732 sampled
Replay capacity (decision transitions)	1,060	60,000	60,000
Batch size x length	16 x 50	16 x 50	16 x 50
Train every / updates	1,000 / 100	1,000 / 100	1,000 / 100
Initial comparison	short-run snapshot	formal pre-update snapshot	formal pre-update snapshot
Final Dreamer mean return	115.463	181.701	322.084

Table 3: Evaluation on ten episodes for each policy, using environment seeds 1000–1009.

Policy	Mean return	Episode sample std.
Random	55.5404	42.5973
Training-initial	70.6835	3.1504
Final Dreamer	181.7011	27.5705

4 Baseline and Experimental Iteration

We first tried the historical TensorFlow route, but the available Python/Keras environment exposed old API and MuJoCo dependency failures. We then used the pinned PyTorch source in the already working Kaggle PyTorch environment. A fresh child process passed the CPU `dm-control` import/reset probe, and the GPU path passed CUDA, EGL pixel rendering, and the short Dreamer update gate. We reused the validated `dm-control -no-deps` route rather than changing the algorithm to fit a new dependency stack. These environment changes matter for reproducibility, but they are not separate scientific experiments.

Before the longer run, we corrected an intermediate configuration with `train_every=1` and `train_steps=8`; the formal run restored the upstream-style schedule of 1,000 decisions and 100 updates. We also increased replay capacity from the short pilot's 1,060 decision transitions to 60,000. The initial comparison is a snapshot of the formal training model after prefill and before its first optimizer update, not a separately initialized policy.

5 Results

The ten-update gate changed the world model, actor, and critic by maximum absolute parameter differences of 0.011696, 0.000710, and 0.000771. The 100k baseline completed 50,000 decisions, 100,000 environment steps, and 4,900 optimizer updates. The independent overnight run sampled 165,366 decisions, 330,732 environment steps, and 16,500 optimizer updates before its 210-minute training limit. Its reported 322.084 evaluation used the earlier saved `params.pkl` at `itr=164999`; the snapshot counter represents 165,000 decisions / 330,000 environment steps, while the larger count is the run's total interaction count. Its train, evaluation, and export subprocesses returned code 0; the downloaded bundle passed a ZIP integrity test and contains the formal initial and final checkpoints, progress, logs, evaluation CSV, and three videos. We keep the two seed-0 runs separate and do not concatenate their curves.

We evaluated three policies on the same ten environment seeds, 1000–1009. Each policy therefore has ten episodes; these are not ten training seeds. Training seed 0 is the only training seed in this report. The reported $\pm$ values are sample standard deviations across evaluation episodes. We controlled the environment seed, but we do not claim that every source of evaluation randomness was fixed.

The final mean return is higher than the training-initial snapshot and random control in both completed runs. The overnight run is higher than the 100k baseline, but the runs differ in budget and wall-clock history and are not a controlled single-factor comparison. The theoretical maximum episodic return is 1,000, so 322.1 is still a limited absolute score. Return is not success rate. The overnight final episodes are relatively close to one another, but this does not establish stability across training seeds.

Figure 2 keeps the independent runs separate. In each left-hand panel, the plotted value is rlpyt's `ReturnAverage`: an average over up to the 100 most recently completed training episodes at each logging point. It is a windowed training diagnostic, not an evaluation learning curve. The 100k

Table 4: Evaluation for the independent overnight run on environment seeds 1000–1009.

Policy	Mean return	Episode sample std.
Random	45.6311	37.4796
Training-initial	73.7413	2.3887
Final Dreamer	322.0840	9.6868

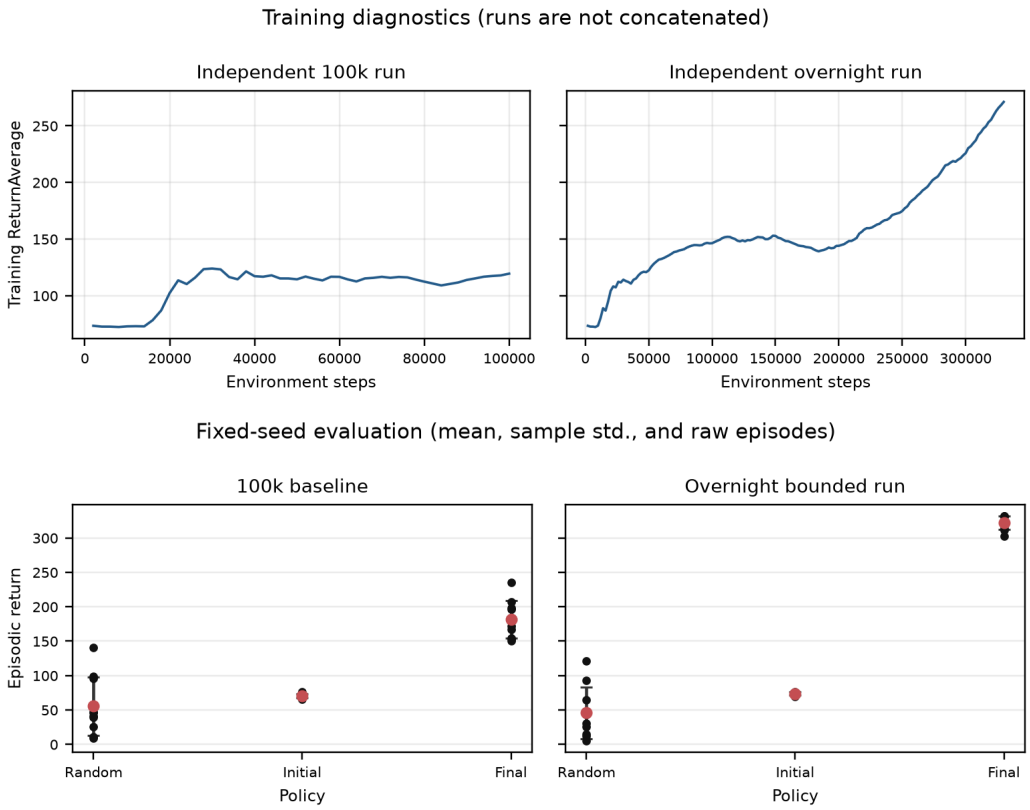


Figure 2: Independent-run diagnostics. The training panels use environment steps on the x-axis and show the windowed rlpyt training return. The evaluation panels show raw returns for the ten fixed environment seeds, with red means and error bars for the sample standard deviation. The 100k and overnight runs are not concatenated.

curve ends near 119, while the overnight curve rises from roughly 73 near prefill to roughly 271 at its last logged point before the actual final counter. Training uses exploration noise, so these curves should not be read as the same quantity as the evaluation tables. The right-hand panels show all raw evaluation episode returns as points, together with the mean and sample standard deviation.

6 Limitations and Future Work

The current evidence comes from one task and one training seed, even though we now have two independent seed-0 runs. The overnight run did not reach its 500k-environment-step target; it stopped at the wall-clock limit. We do not claim statistical significance, stable task mastery, or paper-level performance. The plots do not identify a specific failure behavior, and we do not infer that low returns are caused by a particular code bug or only by insufficient training. The checkpoints were loaded for evaluation in fresh Kaggle child processes, but full restoration of optimizer, replay, and environment state remains untested.

Our next steps follow the proposal. First, we will inspect evaluation videos, the fixed-seed protocol, and the training curve to decide whether the baseline should run longer. Second, we will add training

seeds 1 and 2 with the same budget and configuration. Third, we will run matched-budget sensitivity experiments for the proposal's two directions: (i) imagination horizons such as 5, 15, and 30, which are candidate settings and have not been executed; and (ii) a 4-pixel random-shift image augmentation. For the random-shift study, we will specify padding and crop behavior before implementation, keep the transform consistent within each replay sequence, and document how the encoder input and decoder reconstruction target are transformed. We will not assume that augmentation improves performance.

7 Conclusion

We now have a working Dreamer V1 training and evaluation pipeline for pixel-based `cartpole_swingup`. The formal runs changed all three learned modules, and both final policies scored above their same-run pre-update snapshots under the shared ten-seed evaluation protocol. The overnight run provides evidence that the pipeline can run substantially longer and persist its outputs, but it is still one training seed and below the planned 500k environment-step budget. Replication still requires matched seeds, protocol checks, and the two proposal experiments.

Team Contributions (Pending Confirmation)

Chevy Yang and Zhiyuan Yang both contributed to the seed-0 Dreamer V1 baseline, including experiment setup, execution, and review of the resulting evidence. The report treats this as one shared training run, not as independent training seeds. The finer division of implementation, artifact preparation, and analysis will be finalized after the team confirms the remaining details.

GitHub

Project repository: `https://github.com/DDaniel-code/dreamer-pixels`. The repository records the selected source commit, configuration differences, raw-artifact manifest, and report materials.

References

[1] Julius Frost. Dreamer pytorch. GitHub repository, n.d. URL `https://github.com/juliusfrost/dreamer-pytorch`. MIT-licensed source; fixed commit is recorded in the report text.

[2] Google Research. Dreamer: Learning behaviors by latent imagination. GitHub repository, n.d. URL `https://github.com/google-research/dreamer`. Historical Dreamer V1 reference implementation.

[3] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. *arXiv preprint arXiv:1912.01603*, 2019. URL `https://arxiv.org/abs/1912.01603`. 2019 arXiv preprint; ICLR 2020 conference version.

[4] Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97, pages 2555–2565. PMLR, 2019. URL `https://proceedings.mlr.press/v97/hafner19a.html`.

[5] Yuval Tassa, Yotam Doron, Alistair Muldal, Tom Erez, Yazhe Li, Diego de Las Casas, David Budden, Abbas Abdolmaleki, Josh Merel, Andrew Lefrancq, Timothy Lillicrap, and Martin Riedmiller. Deepmind control suite. *arXiv preprint arXiv:1801.00690*, 2018. URL `https://arxiv.org/abs/1801.00690`.